

"I Am There But...": Remote Students' Experiences of Inclusion through Telepresence Robots

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Abstract

The use of telepresence robots in educational settings offers a promising solution for students who are unable to attend classes in person due to illness, disability, or other barriers. However, the extent to which these technologies facilitate the perceived inclusion of remote students remains underexplored. This study investigates the experiences of remote students using telepresence robots to attend university classes, focusing on factors that mediate their sense of inclusion. Through an inductive analysis of interviews and observations from 22 remote students, we identify technological and socio-pedagogical factors that influence inclusion. Our findings reveal that while telepresence robots enhance physical presence and agency compared to traditional video conferencing tools, challenges such as technological limitations, social norms, and pedagogical practices can hinder full integration. We offer recommendations for educators and institutions to better support remote students and suggest design improvements for telepresence technologies to foster inclusivity in classroom settings.

CCS Concepts

• **Computer systems organization** → **Robotics**; • **Human-centered computing** → **Empirical studies in collaborative and social computing**; **Collaborative and social computing design and evaluation methods**.

Keywords

Robotic telepresence, robot-mediated communication, remote participation, classroom, inclusivity, inclusion

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1 Introduction

Inclusion stands at the forefront of contemporary educational debates, especially as we grapple with issues that prevent physical presence in schools. Whether due to illness, disability, or the recent upsurge in COVID-19 cases, an increasing number of students face extended absences, risking academic lag and feelings of isolation. This emotional disconnect has significant repercussions on both mental well-being and educational outcomes [19, 23]. While the rise of online pedagogies and video conferencing offers academic solutions, they often falter in replicating the vital social dynamics of the traditional classroom—a linchpin of holistic learning experiences.

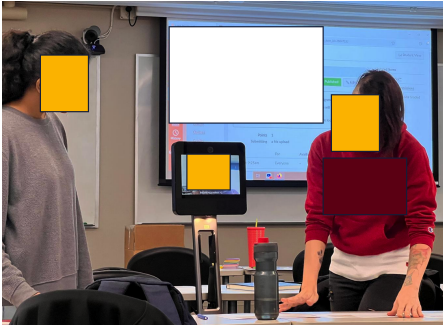
As an answer to this challenge, robotic telepresence emerges as a potentially game-changing medium. This technology offers richer communicative cues such as vocal nuances, spatial mobility, and in some cases, adjustable viewpoints—elements often missing from conventional remote communication tools. However, inherent limitations persist. Users navigate a learning curve in maneuvering these robots, interpreting robot-proxied body language, and presenting themselves through a standardized robot body [36]. People interacting with the user through the robot also have to explore new “rules of engagement” [20].

While existing literature has explored the utility of robotic telepresence across domains—from offices and healthcare to homes—the focus within educational settings remains on facets like user acceptance, interaction quality, and design implications [2, 12, 13, 17, 21, 22, 26]. There is still a knowledge gap regarding how these technologies are perceived as inclusive by remote students from their own perspectives and what mediates such inclusion.

Our motivation for examining inclusion in the use of telepresence robots in the classroom stems from the fact that these technologies were initially designed primarily for office contexts. Yet, interactive activities (e.g., lectures, group work) and the social context (e.g., roles, hierarchies) significantly differ in educational environments. Moreover, telepresence robots can be cognitively taxing to use in parallel with attending to class demands [36]. Thus, it is vital to assess whether these robots adequately support remote students' inclusion and to propose strategies for improvement.

Our research questions center on identifying and describing the factors that hamper or facilitate inclusion from the remote students perspective. We adopt an inductive approach, analyzing screen recordings and interview transcripts from 22 remote students to derive emergent concepts of inclusion. We conclude with recommendations for educators, institutions, and designers to support

Figure 1: Remote and In-Person Students Collaborating on a Classroom Assignment



more inclusive use of mobile telepresence technology in classroom contexts.

2 Related Work

We examine existing research on robotic telepresence, focusing on its role in education and virtual inclusion, to highlight our contribution in this domain.

2.1 Robotic Telepresence and education

Robotic telepresence allows an individual to be both socially and physically present in a distant environment [3, 16]. Mobile telepresence solutions, like Beam, Double, and GoBe, are more than just video conferencing tools; they provide physical embodiment that enables remote users (or “operators”) to move around and engage more fully in a distant environment [16]. While platforms like Zoom and Microsoft Teams address aspects of remote communication, telepresence robots can offer more dynamic and embodied interactions [14, 33].

Researchers have explored the use of telepresence in workplaces [4, 29, 31, 36–38], conferences [11], hospitals [28], museums [10], homes [6, 25, 39–41], and elder care [1, 8, 18]. CHI and CSCW research has highlighted user experience [6, 30, 39], embodiment [32, 33, 38], collaboration support [5], social norms [20], and design recommendations [30, 39].

In educational contexts, robotic telepresence can be particularly beneficial for students who cannot physically attend classes, offering advantages over typical videoconferencing tools [12, 27]. Existing studies have explored aspects such as user acceptance [22], interaction patterns [34], and device personalization [13]. However, how these technologies impact the sense of inclusion and belonging from the remote students’ perspective remains an open question.

2.2 Virtual Inclusion in the Classroom

Virtual inclusion entails creating a flexible, accessible, and interactive environment in virtual or hybrid learning contexts [15]. A few studies focused on telepresence robots focused on inclusion in the context of the classroom [9, 27, 35]. These studies have predominantly focused on K-12 educational settings, examining the use of telepresence robots to support homebound children or patients [9, 27, 35]. Sheehy and Green [35] explored the potential

of mobile telepresence robots in promoting inclusive educational practices. They reported on a small-scale experiment in which groups of children, including those with severe learning difficulties, used telepresence robots to “visit” and explore each other’s classrooms. Their findings suggested that while telepresence robots offer promising opportunities for enhancing social interactions and fostering inclusivity, the current technology is not yet robust enough to support reliable, everyday use in educational settings. Newhart et al. [27] carried out an exploratory case study involving K-12 students, teachers, and administrators to investigate the implementation of telepresence robots for chronically ill students in mainstream classrooms. Their study revealed that telepresence robots fostered a sense of belonging and reduced feelings of isolation among homebound students, while also identifying challenges such as technical limitations and instances of peer bullying that could hinder the effectiveness of virtual inclusion. Last not least, Charteris et al. [9] developed a framework and heuristic for using telepresence robots to enhance virtual inclusion in schools, addressing technological, physical, social, and pedagogical factors. Based on an extensive literature review, they synthesized existing research, including studies on children with severe learning difficulties, to identify key elements for successful inclusion.

In contrast, our research targets higher education, specifically university students. This focus on adult learners in tertiary education highlights a significant gap in the literature, as the unique social dynamics and academic demands of university classrooms necessitate distinct technological and socio-pedagogical considerations for fostering inclusion. While K-12 studies emphasize foundational aspects like social presence and emotional support, higher education environments involve more complex interactions, autonomous learning structures, and diverse pedagogical approaches. By conducting an inductive, in-depth investigation into the factors that mediate or impede inclusion for remote university students using telepresence robots, our study employs ethnographic methods such as participatory observations, surveys, and semi-structured interviews exclusively with remote university attendees. Drawing on participant-defined understandings of inclusion, we explore both technological and socio-pedagogical elements and offer actionable recommendations to enhance the classroom experiences of remote learners. This research extends the discourse on virtual inclusion beyond primary and secondary education contexts, addressing the specific needs of higher education environments and contributing to the development of more effective telepresence solutions for adult learners.

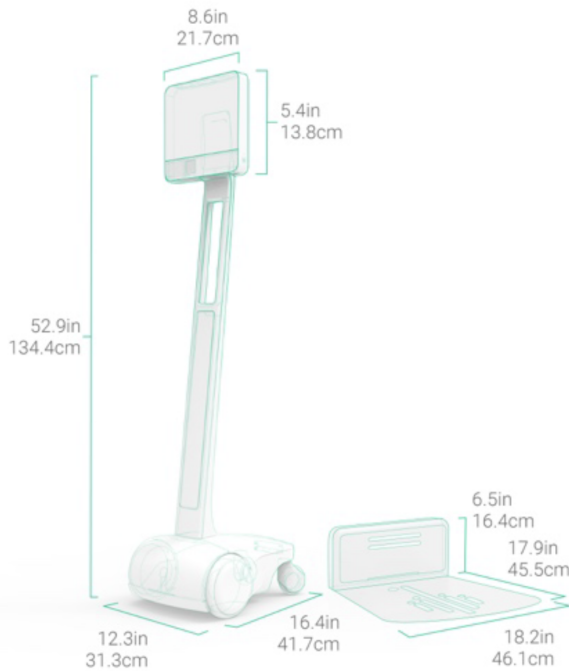
3 Method

We conducted a qualitative field study using ethnographic methods to observe and record how remote students used telepresence robots to attend classes. Our primary data collection methods included participatory observations, surveys, and semi-structured interviews conducted in Fall 2022. This study received IRB approval from Indiana University Bloomington.

3.1 Setting

The study took place in a university building that houses both a research space and a mid-sized classroom. We used two informatics

Figure 2: Diagram with sizes of Beam system and docking station. Photo credit: Blue Ocean Robotics



classes, each with about 45 students, meeting twice a week for 80 minutes over 16 and 13 weeks, respectively. Classes involved lectures, group activities, and student presentations. We deployed three Beam telepresence robots for remote attendance, although at most two were in use simultaneously, with the third as a backup.

3.2 Participants

Forty students in total participated. Twenty-two (11 women, 11 men) served as remote attendees (“drivers”) using telepresence robots at least once and completed surveys and interviews. The remaining 18 classmates responded to a brief survey, and 11 were interviewed. Participants were recruited via in-class announcements and the online learning management system. All participants provided informed consent, and remote attendees received course extra credit (an IRB-approved practice at our institution). We label remote participants as P1, P2, ..., P22, and classmates as PC1, PC2, etc. In this paper we only report from the remote students’ data.

3.3 Apparatus

We utilized three Beam telepresence robots supplied by Blue Ocean Robotics. While a maximum of two Beams were actively used at any given time during the study, the third was available as a contingency measure in case one of the operational Beams encountered technical issues. The Beam robot merges an upright, mobile design with a screen for interactive communication and a wheeled base for easy navigation. It is remotely operated and outfitted with both top and floor cameras, along with an audio system, enabling efficient two-way communication. Control is facilitated through the Beam app.

The robot’s battery typically lasts for two-hour and can be easily parked to charge on the docking station. The Beam pro is smaller and slower than the unit studied at CHI by Neustaedter et al. [24], which might be more recognized among conference attendees. (See figure 2 for a detailed description.)

3.4 Protocols

During each session, one researcher stationed in the lab helped the remote attendee operate the Beam app, start screen recordings, and troubleshoot any issues. Another researcher in the classroom took notes, photos, and brief video clips of classroom interactions. Remote students arrived 30 minutes before class to learn robot navigation and safety, and were encouraged to engage with the class as if they were physically present. After each class session, remote students completed a short survey to provide immediate feedback on their experience. They also scheduled a follow-up interview for a more in-depth exploration of their participation and feelings of inclusion. Additionally, classmates were invited to participate by completing surveys and volunteering for interviews at the end of the study period. The follow-up interviews involved reviewing selected segments of the recorded sessions that researchers identified as particularly noteworthy. Participants were asked to comment on these segments to elaborate on their experiences and perceptions of virtual inclusion. Key interview questions focused on inclusion included: “Can you describe your experience of being included in classroom activities while using the telepresence robot?”, “To what extent did you feel part of the classroom community during the session?”, “What aspects of using the telepresence robot contributed to your feeling of inclusion or exclusion in the classroom?” among other questions that relate to the overall experience of using telepresence robots. This paper primarily reports on the interview data, which provided rich, comprehensive insights into both the breadth and depth of virtual inclusion experiences. The survey and observation data were utilized as a foundation for formulating interview questions, enabling a nuanced understanding of how telepresence robots influence inclusion and participation in university-level classrooms.

3.5 Data Analysis

We used a thematic analysis approach [7] to code screen recordings and interviews. Initial coding involved five researchers independently analyzing six interview transcripts to extract key concepts (e.g., audio-visual quality, usability, interaction, social norms). Through iterative discussions, we consolidated these concepts into broader themes. We then applied this coding scheme to the entire dataset, refining categories and subthemes. We ultimately identified two overarching factors—technological features and socio-pedagogical dynamics—that most significantly shaped remote students’ experiences of inclusion.

4 Findings

Our analysis reveals that remote students’ sense of inclusion is shaped by two primary factors: (1) Technological aspects of the telepresence robots and (2) Socio-pedagogical dynamics within the classroom. Each factor comprises elements that could either

facilitate or hinder the perceived inclusion of remote attendees. Below, we elaborate on these factors.

4.1 Technological Aspects

Telepresence robots serve as the physical and social “avatars” for remote students, mediating their interactions and presence in the classroom. The technological features and limitations of these robots directly impact how included remote students feel within the educational environment.

4.1.1 Technological Features and Capabilities. Participants unanimously acknowledged that telepresence robots offer significant advantages over traditional video conferencing tools, like Zoom and Teams. Their ability to move around the classroom and maintain a consistent audio-video channel helped the remote student appear more “present.” For example, P11 highlighted the enhanced visibility and engagement:

“It’s obvious that I’m physically there. I can move around, raise my hand, and interact and be involved. With Zoom, you can’t tell if someone is engaged because often cameras are off, or they’re just a tiny box in the corner.”

This ability to move and interact mimics in-person participation, making remote students feel more like integral members of the class.

Similarly, P4 emphasized the increased agency and visibility: *“Sometimes the instructor forgets the Zoom folks, but with the robot, I positioned myself near the screen or speaker so I could see and be seen. It gave me more freedom.”* By having the autonomy to position themselves strategically, remote students can better engage with the class, fostering a stronger sense of belonging.

P9 also noted the importance of movement in feeling embedded:

“I feel like if I can attract attention through movement, I feel more embedded in the classroom, I can talk to my classmates and join group discussions. This is hard through Zoom because you can’t move and no one knows you are there or who are you?. So you don’t really feel part of the classroom and you end up disabling camera and sleeping.”

This sentiment underscores how mobility features of telepresence robots contribute to remote students’ inclusion by enabling active participation and visibility.

These insights demonstrate that the enhanced social and physical presence provided by telepresence robots significantly contributes to remote students feeling included and valued within the classroom.

4.1.2 Technological Barriers. Despite the advantages, several technological limitations impede remote students’ sense of inclusion, leading to feelings of marginalization and disconnection. Latency and audio-video glitches were recurrent issues that disrupted the flow of interaction, undermining inclusion. P11 shared: *“There’d be a lag. They [group members] would say, ‘What do you think?’ but by the time I responded, they had moved on. It felt awkward.”* Such delays not only disrupt conversations but also make remote students feel excluded from real-time discussions. Additionally, technical disconnections exacerbated feelings of being undervalued. P11 adds:

“When I got disconnected multiple times and could not hear my classmates, I felt they cared less about my opinion or my answers during the discussion.” This perception directly diminishes their sense of inclusion, as repeated technical failures lead to the perceived lack of consideration of the remote student participation.

Another significant barrier was the difficulty in attracting attention. Eighteen of the 22 remote participants reported feeling overlooked. P7 expressed:

“I was right there in the front, but people would sometimes forget I was actually a person controlling the robot. I wished there was a better way to raise my hand or wave... I wish people remembered I was still there so they talked to me more.”

The inability to effectively signal presence and participation, and project identity led to feelings of invisibility and exclusion from spontaneous interactions.

Physical limitations of the robots further hindered inclusion. The slower mobility compared to human movement made it challenging for remote students to navigate the classroom swiftly, especially during group activities. P20 recounted: *“When class ended, everyone left in a group, but I was stuck alone because the Beam moves slowly. It felt isolating.”* This isolation post-class highlights how technological constraints can lead to social exclusion, reinforcing feelings of being left out.

Overall, while telepresence robots enhance presence, their technological shortcomings can significantly undermine remote students’ sense of inclusion by limiting real-time interaction, visibility, and social integration.

4.2 Socio-Pedagogical Dynamics

Beyond technological factors, the classroom’s social and pedagogical environment plays a crucial role in shaping remote students’ inclusion. These dynamics encompass social norms, learning engagement, and pedagogical adjustments that either support or hinder the integration of remote attendees.

4.2.1 Social Norms and Expectations. Remote students often had to navigate and adapt to existing classroom norms, which were primarily designed for in-person interactions. This adaptation process affected their sense of inclusion, sometimes leading to social marginalization.

P10 described the challenge of maintaining appropriate behavior without disrupting the class:

“The lecture had started, I wanted to move closer to the projected screen to see better and be able to discuss later on. But I didn’t want to roll around and distract people. But I couldn’t hear anything. Eventually, I decided to move, but it felt awkward.”

This hesitation to engage fully due to fear of disrupting the class underscores how unadapted social norms can exclude remote students from active participation. Similarly, P3 chose to remain in a corner to avoid interrupting, which limited their interactions: *“I stayed in the corner for the entire class because I didn’t want to disrupt the flow. It made me feel like an outsider.”* Such behaviors, driven by restrictive social expectations, directly contribute to feelings of exclusion and hinder inclusion. While P10 chose to move, P3, P13

and P16 mentioned they could not move. We also observed that the majority of participants remained in a single location for the entire session, even though their placement did not seem to provide a convenient view of the slides. These experiences highlight that without adapted social norms that account for telepresence, remote students struggle to engage fully, thereby diminishing their sense of inclusion.

4.2.2 Learning Engagement and Pedagogical Adjustments. The degree to which classroom activities were compatible with telepresence significantly influenced remote students' engagement and inclusion.

Group work and hands-on activities posed particular challenges. P17 reflected on the difficulties of participating in a physical project remotely:

"We had an Arduino project in class. My teammates tried to show me what they were doing, but it's impossible to handle a physical board through a robot. I couldn't fully participate."

This inability to engage with tangible materials limited their active contribution, leading to feelings of exclusion from collaborative learning experiences.

Conversely, proactive support from classmates could mitigate these challenges. Many participants noted that when peers made efforts to include them, their sense of inclusion improved. For instance, participants observed classmates moving the robot to facilitate interaction and providing detailed explanations during hands-on activities. This collaborative behavior fostered a more inclusive environment, demonstrating that peer support is pivotal in enhancing remote students' inclusion.

Nonetheless, the reliance on classmates for inclusion indicates that without structured support, remote students remain vulnerable to exclusion during interactive tasks.

Instructor-led pedagogical adjustments were critical in fostering the inclusion of remote students. Deliberate strategies by educators directly influenced how valued and integrated remote attendees felt.

P7 highlighted the importance of instructor check-ins:

"She'd look at the robot and say, 'Do you have a question? Can you see okay?' I felt acknowledged. That makes a big difference because you can't wave your hand as obviously as in person."

Such proactive engagement by instructors reassured remote students of their inclusion, making them feel recognized and supported.

Additionally, distributing materials digitally and assigning remote students to specific groups enhanced their participation:

"The professor asked the AI to send the exercise sheet to the remote student using the robots via email. I would have been hard to participate in the exercise if I did not have it and relied on my classmates."

By ensuring remote students had access to necessary resources and structured roles, instructors facilitated their active involvement, thereby reinforcing their sense of inclusion.

These pedagogical interventions demonstrate that intentional teaching practices are essential for integrating remote students,

highlighting the need for educators to adapt their methods to support inclusive participation.

5 Discussion

5.1 Discussion with prior research

Our findings demonstrate how technological and socio-pedagogical factors jointly influence remote students' experiences of inclusion. Consistent with Newhart et al. [27], we observed that telepresence robots significantly enhance social interactions among remote students, fostering a sense of inclusion and belonging within the classroom. This aligns with our observation that robots facilitate a more integrated presence compared to remote students' perception of traditional video conferencing tools. Additionally, Sheehy and Green [35] emphasized the critical role of teacher support and classroom acclimatization in effectively integrating telepresence robots, which corresponds with our finding that pedagogical adjustments are essential for fostering inclusion.

However, both Newhart et al. [27] and Sheehy and Green [35] also identified several challenges associated with telepresence robot use. While Newhart et al. [27] highlighted issues such as peer bullying, our study revealed that classmates play a crucial role in supporting the inclusion of remote students during hands-on activities, thereby mitigating some of these negative experiences. Similarly, akin to Sheehy and Green [35], we observed significant technological barriers, including limited robot mobility and audio-visual quality, which can lead to remote students feeling marginalized.

Furthermore, many of our findings are reflected in the framework proposed by Charteris et al. [9], which was developed through an extensive literature review. Nevertheless, our study extends these findings by uncovering additional complexities inherent in higher education settings. Unlike the K-12 context, university classrooms involve more autonomous and self-directed learning, complex group dynamics, and diverse pedagogical approaches. We identified different technological barriers, such as limited robot mobility, that specifically impact interactions in higher education environments. The socio-pedagogical dynamics in university settings, including the need for structured group work and the adaptation of teaching strategies to accommodate remote students, present distinct challenges that are not extensively addressed in K-12-focused studies.

5.2 Implications for Educators

Educators play a pivotal role in ensuring telepresence robots serve as genuine conduits for inclusion. They can:

- **Adopt explicit attention signals:** Encourage remote students to use text chat, a "digital hand-raise," or a robot gesture (if available) to signal questions, reducing the risk of being overlooked.
- **Provide scaffolding for group work:** Offer digital access to materials and designate roles for remote students so they can actively participate in hands-on tasks.
- **Incorporate regular check-ins:** Periodic eye contact or direct inquiries ("Any questions from the Beam?") reassure remote students that they are integral to class.

5.3 Design Recommendations

From a design standpoint, several improvements could bolster the effectiveness of telepresence robots in education:

- **Enhanced mobility and speed:** More agile, faster robots or intelligent pathfinding would help remote students keep pace with in-person classmates.
- **Improved attention cues:** Features like a flashing light or gentle chime could help remote students “raise a hand” to speak.
- **Better audio-visual quality:** Noise-canceling microphones and high-resolution cameras reduce the friction of hearing and being heard, especially in crowded or noisy classrooms.
- **Multi-user interactions:** Future telepresence systems might integrate collaborative tools (e.g., shared screens or digital markers) for group assignments.

5.4 Limitations and Future Work

Our study centers on two mid-sized university classes and Beams; findings may differ in other educational levels or larger lecture halls. Additionally, our focus was on short-term use; longer studies could examine changing perceptions and classroom dynamics over a full semester or year. Future work might also compare multiple telepresence platforms, evaluate specific design features, or investigate telepresence in diverse cultural or disciplinary settings.

6 Conclusion

Telepresence robots can expand access for students who cannot attend classes in person, offering a form of mobile, embodied presence that transcends traditional videoconferencing. Through interviews, observations, and screen recordings with 22 remote students, we found that meaningful inclusion requires not only robust technological infrastructure but also supportive social and pedagogical environments. By addressing technological gaps—such as limited mobility and audio-visual glitches—and by adapting classroom norms, group interactions, and teaching strategies, educators can better integrate remote students into daily classroom life. Our findings illuminate the interplay between robots, human actors, and educational contexts, guiding both design and pedagogical practices toward more inclusive futures for remote learners.

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